EVAN ALEKSEYEV

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AI Systems Engineer experienced in training, deployment, and application of CV pipelines to edge hardware

Education

San Jose State University — San Jose, CA
Bachelor of Science in Computer Engineering — *3.7 GPA*

Technical Skills

Languages: Python, C++, C, x86/ARM Assembly, CUDA, Golang (basic)
AI/ML: LangGraph, PyTorch, OpenVINO, ONNX, TensorRT, Quantization, OpenCV, YOLO-*, SAM, CLIP, DINOv2
Infrastructure & Data: Docker, Linux, Git, OpenAPI, FFMpeg, PostgreSQL, SQLite, LanceDB, Grafana Loki
LLM Tooling and NLP: OpenAI-Compatible APIs, Llama.cpp, BERT, edge-optimized VLM inference

Experience

Software Engineer — March 2026 – Present
DeGirum Corp. — *Santa Clara, CA*

- Built production vehicle ReID pipeline (YOLO + OSNet + PaddleOCR) on live RTSP streams with proximity-zone alerting; deployed and benchmarked on DeGirum Orca1 edge accelerator
- Developed multi-camera face recognition attendance system using ArcFace embeddings + LanceDB vector store; designed for production access control and retail tracking use cases
- Led R&D evaluation of VLMs (Qwen3.5, SmolVLM2, Gemma4 E2B, LLaVA, InternVL3.5) for retail anomaly detection; evaluated performance of Vulkan, OpenVINO, CPU, and BLAS backends on Llama.cpp; optimized lightweight VLM serving achieving 700ms TTFT on laptop CPU

Mechatronics Engineer Intern — Sep 2025 – March 2026
Beagle Technology, Inc — *Dublin, CA*

- Co-designed CV pipeline for RealSense camera-based depth and height estimation of grapevines to run on Jetson Orin Nano hardware; emphasis on robust edge-case handling and temporal filtering
- Sole firmware owner (C++): rewrote I/O PCB codebase supporting 4 PCB revisions, 2 machine types, 12-channel relay + 5-channel proportional valve; designed CAN bus communication layer to NVIDIA Jetson
- Contributed to R&D and field deployment of prototype agricultural robotics products, shipping computer vision improvements and PID tuning directly to automated machinery at customer sites

Academic Projects

Texas Instruments Co-Op — August 2025 – May 2026
Texas Instruments — *Santa Clara, CA*

- Engineering lead and scrum master in team of 4; architected multi-node radar fusion pipeline normalizing IWR6843AOP track data (position, velocity, SNR) across 3 distributed nodes into JSON packets for cloud transmission
- Created main system design, chose hardware, and laid out structure for software codebase and hardware imaging processes

Hobby Projects

TensorRTWrapper: C++ TensorRT Inference Engine | *C++, TensorRT, CUDA, ONNX* — 2026

- C++ wrapper around the TensorRT runtime abstracting model deserialization, engine lifecycle, and async job queuing; enables production inference services to submit work without managing low-level TensorRT details

Add Spice: GoPro Video Analysis | *Python, PyTorch, OpenCV, Transformers* — 2025

- Designed DinoV2 backbone + temporal transformer + dense head (RankNet loss) to rate action video intensity 0–10; built end-to-end annotation environment and training pipeline with augmentation and checkpointing

ShakespeareLM: Transformer Language Model | *Python, PyTorch, Transformers* — 2025

- Implemented GPT-2 style decoder-based language model in PyTorch; trained on Shakespeare, Kafka, and Dostoevsky corpuses with custom tokenizer and generation loop. Model learned grammatical and punctuation convention and achieved basic generalization